(i) $du = x\cos x\,dx$

$$\begin{aligned}\int \frac{x}{x\tan x + 1}\,dx &= \int \frac{x\cos x}{x\sin x + \cos x}\,dx \\ &= \int \frac{1}{u}\,du \\ &= \ln|x\sin x + \cos x| + C\end{aligned}$$

Let $u = x\cos x - \sin x,\ du = -x\sin x\,dx$

$$\begin{aligned}\int \frac{x}{x\cot x - 1}\,dx &= \int \frac{x\sin x}{x\cos x - \sin x}\,dx \\ &= -\int \frac{1}{u}\,du \\ &= -\ln|x\cos x - \sin x| + C\end{aligned}$$

(ii) Let $u = x\sec^2 x - \tan x$

$$du = \sec^2 x + 2x\sec^2 x\tan x - \sec^2 x\,dx = 2x\sec^2 x\tan x\,dx$$

$$\begin{aligned}\int \frac{x\sec^2 x\tan x}{x\sec^2 x - \tan x}\,dx &= \frac{1}{2}\int \frac{1}{u}\,du \\ &= \frac{1}{2}\ln|x\sec^2 x - \tan x| + C\end{aligned}$$

$$\begin{aligned}\int \frac{x\sin x\cos x}{(x - \sin x\cos x)^2}\,dx &= \int \frac{x\sec^2 x\tan x}{(x\sec^2 x - \tan x)^2}\,dx \\ &= \frac{1}{2}\int \frac{1}{u^2}\,du \\ &= -\frac{1}{2u} + C \\ &= -\frac{1}{2(x\sec^2 x - \tan x)} + C\end{aligned}$$

This solution was submitted by **Jon**
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